

## International Refrigeration and Compressor Course 2026

### Project #3: Water-Cooled Chiller for High Efficiency Edge Compute Data Center Facility

During the IRCC your task is designing an air-cooled chiller for a high efficiency edge compute data center facility near Champaign, IL. The cycle architecture for the chiller is entirely up to you and must meet the capacity and energy efficiency requirements as stated below. Details regarding the load and environmental conditions are given below.

#### Subtasks:

##### 1. Cycle design

###### a) Problem specification:

- Vapor compression refrigerant: to be chosen (GWP < 150, A1 or A2L/A3 only with charge analysis)
- Chilled fluid: water to Computer Room Air Handler (CRAH)/in-row units (data hall side)
- Heat rejection: dry cooler with intermediate water/propylene-glycol (PG) loop
- Maximum cooling load: 150 kW ( $\approx$  42 refrigeration tons) sensible-dominated IT (i.e. computing) load
- Minimum cooling load: 30 kW (turn-down at low IT utilization)
- Design chilled-water supply / return: 15 / 21 °C (per ASHRAE TC 9.9 W17 envelope)
- Minimum outdoor temperature: TMY3 weather data (Champaign, IL)
- Maximum outdoor temperature: TMY3 weather data; dry-cooler design at 35 °C DB
- Data-hall white-space footprint: 8 m  $\times$  12 m, single hall,  $\approx$  25 racks
- IT operates 24/7/365; assume near-constant load (occupancy is not the driver)
- Meets energy efficiency requirements (see table below)

###### b) Define the operating conditions

###### c) Compare different working fluid and select the most promising

###### d) Calculation and / or selection of

- Evaporator (chilled water side side)
- Condenser (refrigerant  $\leftrightarrow$  glycol loop)
- Outdoor dry cooler (glycol  $\leftrightarrow$  ambient air), with variable-speed fans
- Free-cooling / waterside-economizer heat exchanger (glycol  $\leftrightarrow$  chilled water)
- Compressor(s)
- Electronic expansion valve and any economizer/vapor-injection valve
- Oil separator and liquid receiver, if required by the chosen architecture
- Piping (diameters only — no lengths, fittings, or supports)

###### e) Discuss the pros and cons of the selected system

###### f) Evaluate seasonal performance of the system – report full load COP, Integrated Part Load Value (IPLV), and estimate the facility Power Usage Effectiveness (PUE).

###### g) Creation of a parts list / BOM.

###### h) Representation of the designed cycle in

- p-h-chart

- Piping & instrument diagram (accordingly to chosen hardware)
- i) Selection of appropriate safety measures
- 2. Step-by-step description of commissioning
- 3. Consideration of
  - a) Pressure losses within the heat exchangers
  - b) Automatic control of compressor – expansion device interaction
  - c) Suggestions of improvement

### Energy Efficiency Constraints:

The design must meet the following efficiency targets. Each is tied to a published standard so calculations can be performed and verified consistently.

| Metric                                       | Required value                                                                         | Reference / basis                                                                                                                                                            |
|----------------------------------------------|----------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Full-load COP (cooling)</b>               | $\geq 3.5$ at AHRI rating conditions                                                   | ASHRAE 90.1-2022 Table 6.8.1-3, air-cooled chiller $\geq 150$ kW; compute at 35 °C condenser entering air, 6.7 °C chilled water leaving temperature (LCHW) per AHRI 550/590. |
| <b>(Integrated Part Load Value ) IPLV.IP</b> | $\geq 5.0$                                                                             | AHRI Standard 550/590-2023 §5.2; weighted COP at 100/75/50/25 % load.                                                                                                        |
| <b>Economizer threshold</b>                  | Activate $\leq 10$ °C outdoor air temperature (OAT), full free cooling $\leq 4$ °C OAT | ASHRAE 90.1 §6.5.1 (waterside economizer); sized for 100 % design load when active.                                                                                          |

Notes on computing each metric:

- **Full-load COP:** compute at AHRI 550/590-2023 standard rating conditions (Leaving chilled water temperature (LCHW) 6.7 °C, condenser entering air 35 °C). For this study report COP at the project-specific design point as well.
- **Integrated part load value (IPLV.IP):** weighted average per AHRI 550/590-2023 §5.2 —  $0.01 \cdot A + 0.42 \cdot B + 0.45 \cdot C + 0.12 \cdot D$  where A, B, C, D are COPs at 100/75/50/25 % load with reduced condenser entering temperatures (35 / 26.7 / 18.3 / 12.8 °C).
- **Power usage effectiveness (PUE):** (IT load + other facility energy loads) / IT load on a design-day basis.
- **Economizer:** required by ASHRAE 90.1-2022 §6.5.1; the dry-cooler loop must be capable of providing 100 % of the design cooling load at the specified outdoor air temperature (OAT), via a glycol-to-CHW plate heat exchanger in parallel with the chiller.

**Note:**

- Pressure losses within the piping are to be neglected, pressure losses in heat exchangers are not.
- Subcooling shall be chosen wisely and justified.
- The temperature differences within the heat exchangers have to be defined and considered in the computation.
- The **essential** part of any used specification sheet has to be attached on the course work documentation (if EES is used, please add a PDF with your EES Code as well)
- The required properties can be calculated (rounded!) with the software
  - CoolProp (<http://www.coolprop.org/>)
- When calculating pipe diameters use Table 1:

**Table 1: Streaming velocities within charge lines**

| <i>state</i>                   | <i>v in (m/s)<sup>1</sup></i> |
|--------------------------------|-------------------------------|
| <i>gaseous – low pressure</i>  | 4.5 – 20                      |
| <i>gaseous – high pressure</i> | 10 – 18                       |
| <i>liquid</i>                  | <1.5                          |

- As a guideline, the Table 2 lists potential manufacturer as well as suppliers for refrigeration components:

**Table 2: Supplier / Manufacturer for selected components**

| <i>Part</i>           | <i>Supplier / Manufacturer</i> | <i>Link</i>                                                                                                                                                   |
|-----------------------|--------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <i>Compressor</i>     | <i>Bitzer</i>                  | <a href="https://www.bitzer.de/websoftware/">https://www.bitzer.de/websoftware/</a>                                                                           |
|                       | <i>GEA Bock</i>                | <a href="https://www.gea.com/en/productgroups/compressors/index.jsp">https://www.gea.com/en/productgroups/compressors/index.jsp</a>                           |
|                       | <i>Emerson</i>                 | <a href="https://climate.emerson.com/en-us/brands/copeland">https://climate.emerson.com/en-us/brands/copeland</a>                                             |
|                       | <i>MAYEKAWA</i>                | <a href="http://www.mayekawa.com/products/compressors/">http://www.mayekawa.com/products/compressors/</a>                                                     |
|                       | <i>Sabroe</i>                  | <a href="https://www.sabroe.com/en/products/sales-tools/comp1/">https://www.sabroe.com/en/products/sales-tools/comp1/</a>                                     |
|                       | <i>Danfoss</i>                 | <a href="https://www.danfoss.com/en/products/">https://www.danfoss.com/en/products/</a>                                                                       |
|                       | <i>SECOP</i>                   | <a href="https://www.secop.com/solutions/compressor-qa-tools/compressor-service/">https://www.secop.com/solutions/compressor-qa-tools/compressor-service/</a> |
| <i>Consumables</i>    | <i>ITE Sint</i>                | <a href="http://www.ite-tools.com">http://www.ite-tools.com</a>                                                                                               |
|                       | <i>ESK Schultze</i>            | <a href="http://www.esk-schultze.de/">http://www.esk-schultze.de/</a>                                                                                         |
|                       | <i>Danfoss A/S</i>             | <a href="https://www.danfoss.com/en/products/">https://www.danfoss.com/en/products/</a>                                                                       |
|                       | <i>CARLY</i>                   | <a href="http://www.carly-sa.com/">http://www.carly-sa.com/</a>                                                                                               |
| <i>Heat Exchanger</i> | <i>Kelvion</i>                 | <a href="https://www.kelvion.com/us/products/">https://www.kelvion.com/us/products/</a>                                                                       |
|                       | <i>Güntner</i>                 | <a href="http://www.guntnerus.com/products/">http://www.guntnerus.com/products/</a>                                                                           |
|                       | <i>SWEP</i>                    | <a href="https://www.swepusa.com/">https://www.swepusa.com/</a>                                                                                               |
|                       | <i>Alfa Laval</i>              | <a href="https://www.alfalaval.com/products/">https://www.alfalaval.com/products/</a>                                                                         |
|                       | <i>Tranter</i>                 | <a href="https://www.tranter.com/">https://www.tranter.com/</a>                                                                                               |
| <i>Pumps</i>          | <i>WITT</i>                    | <a href="https://www.th-witt.com/en/">https://www.th-witt.com/en/</a>                                                                                         |

<sup>1</sup> These approximated values are a fair tradeoff considering costs, efficiency, streaming noises as well as oil circulation rate (see ASHRAE Handbook, Refrigeration, 2006)

| <i>Part</i> | <i>Supplier /<br/>Manufacturer</i> | <i>Link</i>                                                                                 |
|-------------|------------------------------------|---------------------------------------------------------------------------------------------|
|             | Grundfos                           | <a href="https://www.grundfos.com/products.html">https://www.grundfos.com/products.html</a> |
|             | Wilo                               | <a href="https://wilo.com/us/en_us/">https://wilo.com/us/en_us/</a>                         |

## References:

ASHRAE Standard 90.1-2022, *Energy Standard for Sites and Buildings Except Low-Rise Residential Buildings* — chiller minimum efficiencies (Table 6.8.1-3), economizer requirements (§6.5.1), VSD requirements (§6.4.3.10, §6.5.4).

ASHRAE Standard 90.4-2022, *Energy Standard for Data Centers* — Mechanical Load Component (MLC) and Electrical Loss Component (ELC); climate-zone-specific maxima in Table 6.2.1.2; calculation procedure in Normative Appendix A.

AHRI Standard 550/590-2023 (I-P), *Performance Rating of Water-Chilling and Heat Pump Water-Heating Packages Using the Vapor Compression Cycle* — full-load and IPLV/IP rating procedures and test conditions.

ASHRAE Technical Committee 9.9 (2021), *Thermal Guidelines for Data Processing Environments*, 5th ed. — allowable/recommended inlet-air and chilled-water envelopes (classes A1–A4, W17–W45).

The Green Grid (2014), *White Paper #49: PUE™ — A Comprehensive Examination of the Metric* — definitions of PUE and pPUE.

ASHRAE Handbook — HVAC Applications (2023), Ch. 20 *Data Centers and Telecommunication Facilities* — load characterization, dry-cooler sizing, economizer integration.

U.S. EPA AIM Act Final Rule, Technology Transitions §40 CFR Part 84 (2023) — GWP limits applicable to new chillers (informs refrigerant selection).

**Your project contact is: Prof. Craig Bradshaw, [crbrad@illinois.edu](mailto:crbrad@illinois.edu)**